

Meridian Parts

European service-hub expansion — board decision brief

Recommendation: fund **Czechia + Spain** as a gated option, with €225,000 year-zero capex and 5 FTE, only after the 90-day launch gates confirm operational readiness. Under the board policy's base arithmetic, the option produces €198,099 annual incremental contribution and pays back in 1.14 years; it is not a statistically proven or causal forecast.

Decision at a glance

Metric	Selected option	Strongest alternative	Defer
Base annual contribution	€198,099	€186,329	€0
Year-zero capex	€225,000	€240,000	€0
FTE	5	6	0
Payback	1.14 years	1.29 years	—
Stress annual contribution	€32,180	€29,264	€0

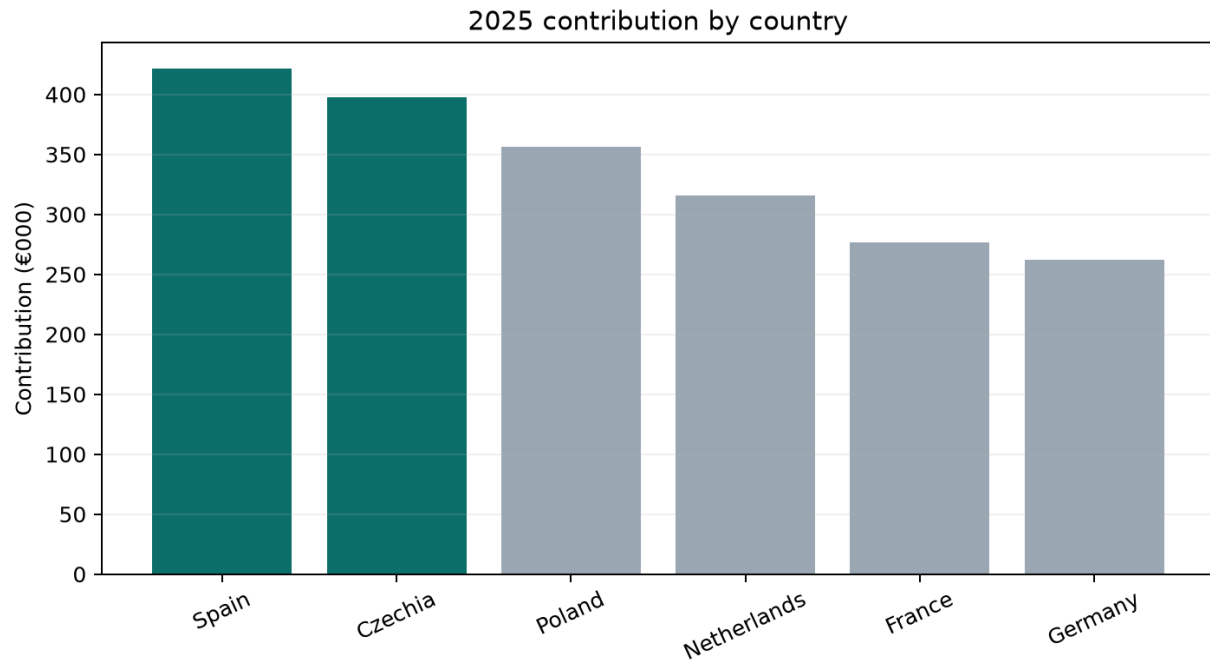
Why this is the decision

The recommendation is driven by the observed 2025 contribution base and the client-supplied policy arithmetic, not by population or GDP as a demand proxy. The selected option is the highest base-case feasible alternative after applying the hard €450,000 capex and 7 FTE limits. Its concession is upfront capital and operating complexity; the strongest alternative (POL+ESP) gives up €11,770 of base annual contribution while requiring €-15,000 different capex. That trade-off is material enough to require a gate, not a blanket commitment.

2025 baseline diagnosis

The reconciled 2025 base contains 1,254 shipped orders and 77,436 units. Gross sales are €4,464,622, refunds €80,831, net sales €4,383,792, net COGS €2,246,919, fulfillment €106,862, and contribution €2,030,010 (46.3% margin). Returned units are 1,081. Booked revenue is gross sales less refunds in the original sale-month translation; cash timing can differ because returns are received through the cutoff, while contribution additionally deducts net COGS and nonrefundable fulfillment. These are not interchangeable measures.

Country	Orders	Units	Net sales	Contribution	Margin	Returned units
Spain	209	13,931	€850,839	€421,900	49.6%	252
Czechia	209	13,733	€815,307	€397,335	48.7%	288
Poland	209	13,163	€761,890	€356,494	46.8%	180
Netherlands	209	12,576	€697,733	€315,644	45.2%	144
France	209	11,880	€632,231	€276,271	43.7%	144
Germany	209	12,153	€625,792	€262,366	41.9%	73



Evidence and accounting controls

The two order extracts, correction file, returns file, unit-cost table, scenario policy and data dictionary are synthetic client inputs. The order pipeline removes exact repeated rows, selects the highest revision per order ID, keeps valid zero-price shipments, excludes cancelled/test/future shipments, and counts distinct order IDs. Returns select the highest revision per return ID, remove exact repeats, add distinct return IDs, require an eligible linked order, and use the inclusive 2026-01-31 cutoff. No orphan return is matched by guess. The original sale month's mean ECB business-day quote is used for both sale and refund conversion; PLN and CZK are not inverted.

- scope: 2025 shipped orders and returns received through inclusive 2026-01-31
- orders_raw_rows: 1328
- orders_unique_after_dedup: 1297
- orders_exact_duplicate_rows_removed: 13
- orders_revision_replacements: 18
- orders_same_revision_conflicts: 0
- orders_eligible: 1254
- orders_excluded_cancelled: 24
- orders_excluded_test: 18
- orders_excluded_not_2025: 1
- valid_zero_price_orders_preserved: 12
- returns_raw_rows: 264
- returns_unique_after_dedup: 243
- returns_exact_duplicate_rows_removed: 20
- returns_revision_replacements: 1
- returns_eligible: 241
- returns_orphan_returns: 1
- returns_on_ineligible_orders: 0
- returns_after_cutoff: 1
- fx_observations_2025: 255
- fx_pln_months: 12
- fx_czk_months: 12
- fx_missing_rate_cells: 0
- missing_fx_rates: 0
- missing_unit_costs: 0
- returns_exceed_shipped_units: 0
- negative_gross_sales: 0
- negative_refunds: 0
- monthly_to_country_reconciliation: True
- handling: Anomalies are retained in reconciled CSVs and counted here; orphans and ineligible/late returns are excluded per data dictionary, never silently matched.

Market context — context, not demand proof

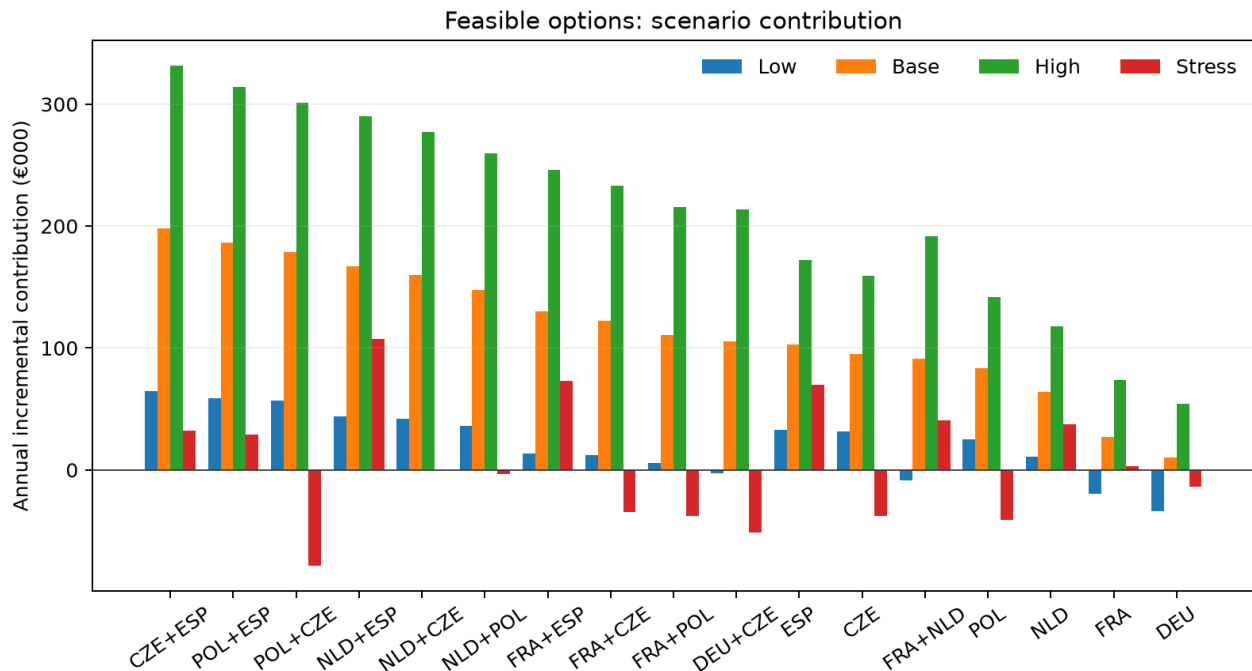
Market	2022–24 population change	2024 population	2024 GDP/person (current US\$)
Germany	0.4%	83,516,593	\$56,104
France	0.5%	68,551,653	\$46,103
Spain	2.2%	48,848,840	\$35,327
Poland	-0.7%	36,559,233	\$25,104
Netherlands	1.7%	17,993,485	\$67,465
Czechia	2.2%	10,905,028	\$31,828

The archived World Bank series show differing scale and recent population trajectories, while current-US-dollar GDP per capita supplies economic context. Neither series is a direct measure of Meridian Parts demand, service density, customer concentration, or hub savings. The source room preserves 2022–2024 values, units, missingness and the 2026-07-13 revision vintage; the ECB archive preserves the reference-rate observations used in the calculation.

Hub economics and trade-offs

The hub policy defines annual incremental contribution as $\text{baseline contribution} \times \text{volume uplift} + \text{projected units} \times (1 + \text{uplift}) \times \text{saving per unit} - \text{annual fixed cost}$. Capex is not subtracted from that annual measure; it is year-zero cash and is used in simple undiscounted payback. The stress case jointly applies a 3% gross-sales refund shock and a 10% net-sales FX shock for PLN/CZK markets. Pairs add country figures without synergy.

Alternative	Low	Base	High	Stress	Capex	FTE	Base payback
CZE+ESP	€64,619	€198,099	€331,580	€32,180	€225,000	5	1.14
POL+ESP	€58,562	€186,329	€314,096	€29,264	€240,000	6	1.29
POL+CZE	€56,856	€178,995	€301,134	€-78,570	€205,000	5	1.15
NLD+ESP	€43,994	€166,931	€289,869	€107,630	€270,000	6	1.62
NLD+CZE	€42,289	€159,598	€276,907	€-203	€235,000	5	1.47
NLD+POL	€36,232	€147,828	€259,423	€-3,119	€250,000	6	1.69
FRA+ESP	€13,687	€129,849	€246,012	€73,105	€340,000	7	2.62
FRA+CZE	€11,981	€122,516	€233,050	€-34,729	€305,000	6	2.49
FRA+POL	€5,924	€110,745	€215,566	€-37,645	€320,000	7	2.89
DEU+CZE	€-2,549	€105,472	€213,492	€-51,410	€375,000	7	3.56
ESP	€33,162	€102,716	€172,270	€70,007	€130,000	3	1.27
CZE	€31,457	€95,383	€159,309	€-37,827	€95,000	2	1.00
FRA+NLD	€-8,644	€91,348	€191,339	€40,721	€350,000	7	3.83
POL	€25,400	€83,612	€141,825	€-40,743	€110,000	3	1.32
NLD	€10,832	€64,215	€117,598	€37,623	€140,000	3	2.18
FRA	€-19,476	€27,133	€73,741	€3,098	€210,000	4	7.74
DEU	€-34,006	€10,089	€54,183	€-13,583	€280,000	5	27.75



Decision conditions and flip points

The base-case choice is supported only as a transparent planning estimate. The selected option's policy break-even volume uplift is approximately 2.7% under unchanged unit savings and fixed costs; below that, the base arithmetic does not cover recurring fixed cost. A second flip condition is operational: if the 90-day gate cannot evidence the uplift/savings prerequisites or if the downside KPI indicates stress-like performance, defer capex and retain the central-hub baseline. The strongest alternative remains relevant because it uses different capital and staffing; do not treat its lower base contribution as a universal ranking outside the stated criteria.

Staged 90-day implementation

Stage / owner	Action and dependency	Decision gate / KPI
Days 0–30 — COO + Finance	Lock baseline definitions, order/return controls, site shortlist, labor and lease diligence; dependency: finance approves a single contribution bridge and source register.	Gate 1: no unresolved data or legal red flags. KPI: $\geq 99.5\%$ order-ID reconciliation; 100% refunds linked or quarantined.
Days 31–60 — Operations + Procurement	Validate 3PL/lease quotes, staffing plan, pick-pack process, service-level design and savings measurement; dependency: site and provider quotes.	Gate 2: all-in capex within budget and staffing within 7 FTE. KPI: quoted cost variance $\leq 10\%$; documented saving €/unit by market.
Days 61–90 — Country lead + IT	Run a controlled launch readiness test using simulated order flow and returns, monitor service levels and FX exposure; dependency: systems, training and inventory controls.	Gate 3: release capex only if evidence supports at least the break-even uplift and no critical control fails. KPI: $\geq 95\%$ on-time dispatch, return processing $\leq 48h$, weekly contribution bridge.

Key risks and mitigations

Demand uplift and savings are planning assumptions rather than observed hub effects; mitigate with staged release and a measured €/unit bridge. Returns could erode savings; monitor return rate, refund EUR and restocked units separately. FX reference rates are analytical translations, not transaction hedge outcomes; retain monthly PLN/CZK monitoring. Capex and labor quotes may drift; set approval tolerances and a stop-work gate. Hub pairs are modeled without synergy; any claimed network benefit is outside this estimate and requires separate evidence.

Limitations, evidence register and reproducibility

Material limits: the client inputs are synthetic; no customer interviews, demand experiment, lease quotes, service-level observations or actual hub pilot are present. The World Bank/ECB material is an archived official snapshot, not a live pull at analysis time. The scenario policy is simple undiscounted arithmetic and does not include tax, working capital, inventory holding, cannibalization, network interaction, ramp timing, or discounting. Returns are known only through the cutoff; later cash timing can differ. Recommendations therefore require gates and should be reopened if the break-even uplift, fixed cost, capex, return rate or FX assumptions change materially.

ID	File	Status	Period / units	Original URL
SR-01	data-dictionary.md	synthetic client input	client accounting policy / see file / not applicable	http://127.0.0.1:49795/data-dictionary.md
SR-02	ecb-history.csv	public archived official snapshot	1999-01-04 to 2026-09-25 observations / local currency units per EUR	https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip
SR-03	ecb-history.zip	public archived official snapshot	official ECB history archive / see file / not applicable	https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip
SR-04	gdp-per-capita.json	public archived official snapshot	2022-2024 / current US\$ per person	https://api.worldbank.org/v2/country/D EU;FRA;NLD;POL;CZE;ESP/indicator/NY.GDP.PCAP.CD?date=2022:2024&format=json&per_page=1000
SR-05	hub-options.csv	synthetic client input	planning assumptions / EUR, FTE, EUR per unit	http://127.0.0.1:49795/hub-options.csv
SR-06	order-corrections.csv	synthetic client input	2025-02-08 to 2025-11-08 corrections / local currency sales; EUR fulfillment	http://127.0.0.1:49795/order-corrections.csv
SR-07	orders-part1.csv	synthetic client input	2025-01-01 to 2026-01-05 shipment records / local currency sales; EUR fulfillment	http://127.0.0.1:49795/orders-part1.csv
SR-08	orders-part2.csv	synthetic client input	2025-01-04 to 2026-01-05 shipment records / local currency sales; EUR fulfillment	http://127.0.0.1:49795/orders-part2.csv
SR-09	population.json	public archived official snapshot	2022-2024 / persons	https://api.worldbank.org/v2/country/D EU;FRA;NLD;POL;CZE;ESP/indicator/SP.POP.TOTL?date=2022:2024&format=json&per_page=1000
SR-10	returns.csv	synthetic client input	2025-01-28 to 2026-02-01 returns / local currency refunds; physical units	http://127.0.0.1:49795/returns.csv
SR-11	scenario-policy.md	synthetic client input	board scenario policy / see file / not applicable	http://127.0.0.1:49795/scenario-policy.md
SR-12	source-register.json	synthetic client input	not applicable / see file / not applicable	http://127.0.0.1:49795/source-register.json
SR-13	unit-costs.csv	synthetic client input	2025 effective-dated costs / EUR per unit	http://127.0.0.1:49795/unit-costs.csv

Reproduce from the workspace: `python3 scripts/build_analysis.py`. Verification outputs include `deliverables/metrics.json`, `deliverables/meridian_parts_analysis.xlsx`, the generated report and presentation, reconciled CSVs in `analysis/`, and hashes in `evidence/source_register.json`.